DEVOPS ENGINEER EXPERT (AZ-400) | CAPSTONE PROJECT

Submission Guide

What to submit, how to name it, how it is marked, and how to hand it in.

Item	Value
Submission deadline	16:45 Hrs - 18 October 2026
Submit to	https://bit.ly/PSEBCorVitDevOps3
Maximum upload size	100 MB
Questions go to	All Participants
Cloud track	Azure, AWS, or both (the trainer tells you which one is marked)

In one sentence

Submit ONE zip file containing a short report, your screenshots named by mission, your written deliverables and a links file, all showing your STUDENT_NAME and your verify token, and only after you have run the cleanup script.

1. What you submit

#	Item	Format	Marks*
1	Cover sheet (Section 3)	Page 1 of the report	required
2	Evidence screenshots, one set per mission (Section 4)	PNG or JPG	85
3	Verify and report output (M10)	PNG	10 (inside the 85 above, see Section 7)
4	Written deliverables D1 to D5 (Section 5)	DOCX, PDF or Markdown	10
5	Cleanup proof: the CLEAN line and the billing page	PNG	5
6	Links file with your GitHub repository, pull request and build numbers	TXT	supports marks
7	Check-question answers (one per mission) inside the report	DOCX or PDF	inside each mission

Marks are shown against the capstone marking scheme of 100 marks (Section 7). Do not submit the scripts themselves; the trainer already has them.

2. Folder and file naming

Create this folder on your laptop, fill it, then zip it. Use your own STUDENT_ID everywhere. No spaces, no capital letters in file names except the mission code.

```
CAPSTONE_<id>_<cloud>/                e.g. CAPSTONE_amk07_azure
  00_cover_sheet.docx                 filled cover sheet (Section 3)
  01_report.docx                      short report: one section per
mission, with answers
  02_evidence/
    M0/  M1/  M2/  M3/  M4/  M5/  M6/  M7/  M8/  M9/  M10/
                                      screenshots named M#_what.png (list
in Section 4)
  03_deliverables/
    D1_stakeholder_update.docx
    D2_change_request.docx
    D3_readme_and_case_study.docx
    D4_freelance_offer.docx
    D5_azure_vs_aws.docx              only if you did both clouds or
worked in a pair
  04_links.txt                        GitHub repo URL, merged PR URL,
Jenkins build numbers
```

- **Zip name:** CAPSTONE_<id>_<cloud>.zip, for example CAPSTONE_amk07_azure.zip. One zip per cloud.
- **Image names:** M5_red.png, M7_pull_error.png (mission code, underscore, short description).
- **Dates and times** in screenshots must be visible. Do not crop out the browser address bar or the shell prompt.

3. Cover sheet

Copy this table to the first page of 00_cover_sheet.docx and fill every row.

Field	Your entry
Full name	
STUDENT_ID	
Cloud used (Azure / AWS / both)	
Region or location used	
Verify score and token (from M10)	Score: ___ / ___ Token: _____
Date and time of verify	
GitHub repository URL	
Cleanup done (date and time)	
I confirm this is my own work and that I removed all cloud resources	Signature or typed name (Recomended): _____

4. Evidence checklist by mission

Tick each item before you zip. A screenshot with the wrong or missing STUDENT_ID, token or timestamp can score zero for that item. If a step does not apply to your cloud, write "not applicable on <cloud>" in the report instead.

Mission	Required evidence	File names
M0 Prepare	Budget page; check output with green [OK] lines	M0_budget.png, M0_check.png
M1 Source and flow	git log --oneline --graph; merged pull request showing your username; your manual time estimate	M1_git_log.png, M1_pr.png, estimate in the report
M2 Network	Firewall rules with your IP; the curl timeout from the shell	M2_rules.png, M2_timeout.png
M3 Identity	Role assignments or IAM role; secret read (length only); denied write; Azure only: lock error	M3_iam.png, M3_read.png, M3_write_denied.png, M3_lock.png
M4 Compute and Ansible	Server list with tags; recap with changed=0; drift run showing changed=1	M4_instances.png, M4_changed0.png, M4_drift.png
M5 Jenkins	Green stage view with the approval step; red build with number and failing line; registry tags	M5_stage_view.png, M5_red.png, M5_tags.png
M6 Containers	Image history; whoami output; cache lines after the code change	M6_history.png, M6_whoami.png, M6_cache.png
M7 Kubernetes	Pods running; browser with id and token; loop of 200 responses; pull error for 9.9; rollback; self-heal; decoded Secret	M7_pods_running.png, M7_browser.png, M7_loop_200.png, M7_pull_error.png, M7_rollback.png, M7_heal.png, M7_secret_decoded.png
M8 Security and IaC	Both scan outputs; your table of five production shortcuts with fixes	M8_scan.png, M8_shortcuts_table.docx
M9 Monitoring	Alert rule; CPU chart; alert fired; e-mail; audit trail	M9_alert_rule.png, M9_cpu_chart.png, M9_alert_fired.png, M9_mail.png, M9_audit.png
M10 Prove and clean	Verify result (score, token, time); report output; the CLEAN line; billing page	M10_verify.png, M10_report.png, M10_clean.png, M10_billing.png

What must never appear in a screenshot

GitHub tokens, the Jenkins unlock password, secret values (only the length is allowed), cloud account keys, and other students' IDs. Blur or crop before saving. Submitting a secret is a rule violation, not just a lost mark.

5. Written deliverables (10 marks)

File	What to write	Length
D1 stakeholder update	Message to a non-technical manager about your red build: headline, business impact, cause in plain words, what you did, new time and owner. Quote your real build number.	120 words maximum
D2 change request	Friday 4:40 pm scenario: decision in one sentence, three reasons, reply message, controls that would allow a safe yes, and which controls you actually have in the lab. Use a real build number or commit hash.	1 page; message 120 words maximum
D3 README and case study	README of your capstone-app repository (what it is, architecture picture, how to run, what you learned). One-page STAR case study with numbers from your own run.	README plus 1 page
D4 freelance offer	One-page offer with three packages, your own hour estimates, scope, exclusions, revisions and acceptance test.	1 page
D5 Azure vs AWS	Comparison table from your own results (pairs or both-cloud students only).	1 table

Write in your own words, in plain business language. Generic text without your own numbers, IDs and build numbers receives at most 3 of 10 for that item.

6. Report structure (01_report.docx)

Keep the report short. Screenshots stay in the evidence folder; the report only points to them and answers the questions.

Section	Content
Page 1	Cover sheet
Section per mission (M0 to M10)	Three to six lines: what you did, one thing that surprised you, the file names of your screenshots, and your answer to that mission's check question using your own output.
One real error	The exact error text, the command, minutes lost, the diagnostic steps in order, which step revealed the cause, and the log or alert that would have shown it earlier.
Value stream	Your manual estimate (M1) versus the report result: lead time, wait time, flow efficiency, and one sentence on what the difference means.
Lessons	Three things you would do differently in a real project.

7. How it is marked (100 marks)

Part	Marks	What earns the marks
M0 and M1	10	Check output, repo URL, merged pull request, manual estimate
M2, M3, M4	15	Firewall rules, denied write, changed=0, drift repaired
M5 and M6	20	Green run with approval, red run with failing line, layer cache answer
M7	15	Zero-downtime loop, 9.9 events, rollback, self-heal, answer
M8 and M9	15	Five-shortcuts table, alert fired, audit trail
Verify score	10	Score divided by total, times 10; token and time must match
Deliverables D1 to D4	10	Own words, own numbers, correct Azure or AWS names
Cleanup proof	5	CLEAN line plus billing page

Each written answer is scored out of 10 as: personal evidence 3, correct technical explanation 3, correct Azure or AWS mapping 2, reasoning and honest self-assessment 2. ***The trainer may ask a 2-minute oral follow-up on any answer and may ask you to re-run one command on screen***.

8. Rules

- **Own work.** Evidence must show your STUDENT_ID and your token. Identical evidence in two submissions loses marks for both.
- **Tools.** You may use a tool to polish wording, but the numbers, output and decisions must come from your own run. You must be able to explain any line in the oral follow-up.
- **Cleanup first.** Submit only after the cleanup script printed CLEAN. A submission with servers still running loses the cleanup marks and may be reported.
- **Late work.** Not Accepted
- **Re-submission.** Only till mentioned time - 16:45 Hrs.
- **Problems on the day.** If your account or the cloud failed, submit what you have plus the exact error text and a screenshot, and tell the trainer before the deadline.

9. Final self-check before you zip

Done	Check
[ ]	Folder and zip are named with my STUDENT_ID and cloud
[ ]	Every screenshot is named M#_what and shows my id, token or time where required
[ ]	No token, password, key or secret value is visible anywhere
[ ]	Verify screenshot shows score, token and time
[ ]	Cleanup screenshot shows CLEAN, and billing shows no running charges
[ ]	Report answers all eleven check questions with my own output
[ ]	D1 to D4 are written in my own words, with my build number or commit hash
[ ]	Links file opens: repository, pull request, build numbers
[ ]	Zip opens on another computer and is below the size limit

Common mistakes

- Cropping out the token or the timestamp.
- Pasting the secret value instead of its length.
- Submitting before cleanup, or deleting only the servers and leaving the network.
- Answering with textbook definitions instead of your own output.
- Forgetting to blur the GitHub token.

10. For the trainer: receiving checklist

Step	Action
1	Open the zip; confirm folder and file names follow Section 2.
2	Compare the verify token and time with the class sheet. A mismatch is a flag, not an automatic fail: ask for a live re-run.
3	Check the CLEAN screenshot and, if you have access, the subscription or account for leftover resources tagged with the student id.
4	Mark mission evidence against Section 7; pick one oral follow-up question per student.
5	Record the mark, any flag and the oral result on the class sheet.